

REPORT

Title

Implementation of the Water Jug Problem Using Python

Aim

To implement the Water Jug Problem in Python and determine a sequence of valid operations to obtain the required quantity of water.

Problem Statement

The Water Jug Problem is a state-space search problem in Artificial Intelligence involving two water jugs of fixed capacities and an unlimited water supply. The objective is to measure a specified quantity of water by performing valid operations such as filling, emptying, and transferring water between the jugs. The program should determine and display a valid sequence of operations that results in the desired amount of water.

Requirements

- Python 3.x

Procedure

1. Define the capacities of the two water jugs.
2. Specify the target amount of water to be measured.
3. Represent the initial state with both jugs empty.

4. Generate all possible valid operations.
5. Create new states after each operation.
6. Ignore previously visited states to avoid repetition.
7. Continue exploring states until the target quantity is reached.
8. Display the sequence of states that leads to the solution.

Program

```
from collections import deque

def water_jug(cap1, cap2, target):
    visited = set()
    queue = deque([(0, 0), []])

    while queue:
        (a, b), path = queue.popleft()

        if (a, b) in visited:
            continue
        visited.add((a, b))

        path = path + [(a, b)]

        if a == target or b == target:
            return path

        next_states = [
            (cap1, b),
            (a, cap2),
            (0, b),
            (a, 0),
            (a - min(a, cap2 - b), b + min(a, cap2 - b)),
            (a + min(b, cap1 - a), b - min(b, cap1 - a))
        ]

        for state in next_states:
            if state not in visited:
                queue.append((state, path))

    return None

capacity1 = 4
capacity2 = 3
target = 2
```

```
solution = water_jug(capacity1, capacity2, target)

if solution:
    print("Solution Path:")
    for state in solution:
        print(state)
else:
    print("No Solution Exists")
```

Output

```
Solution Path:
(0, 0)
(0, 3)
(3, 0)
(3, 3)
(4, 2)
```

Result

The Water Jug Problem was successfully implemented in Python. The program generated a valid sequence of operations to obtain the required amount of water using the given jug capacities.

Conclusion

The Water Jug Problem demonstrates **state-space representation** and **search techniques** in Artificial Intelligence. By systematically exploring valid states and avoiding repeated configurations, the algorithm efficiently finds a solution. This problem is widely used to illustrate problem-solving, search strategies, and constraint satisfaction in AI.